Session 20, Deferred Taxes and Valuation Allowances II: Multi Year Calculations, Valuation Allowances and Why Deferred Taxes Matter

By the end of this session you must be able to run a full multi year deferred tax calculation with journal entries (both income statement and balance sheet approaches), know what a valuation allowance is and when it applies, understand why deferred taxes matter more in some countries than others, and be able to connect deferred tax numbers to analytical/valuation questions using the academic evidence covered.

Plan

1. Worked Example 1: Bad Debt Allowance Over Two Years
2. Worked Example 2: Accelerated Tax Depreciation Over Six Years
3. Worked Example 3: Balance Sheet Approach, Multiple Items at Once
4. Valuation Allowances
5. How Important Are Deferred Taxes? It Depends on the Country
6. Why Deferred Taxes Matter for Valuation and Analysis
7. Academic Research Insights
8. Common Mistakes and Exam Traps
9. Must Know Absolutely

1. Worked Example 1: Bad Debt Allowance Over Two Years

Setup: Year 1 revenues €10,000, expenses €7,000 (including a €1,000 bad debt allowance for a client believed to be in financial difficulty, not yet tax deductible until the client is formally written off). Year 2 revenues €12,000, expenses €7,500 (the client is written off in Year 2, at which point the €1,000 becomes tax deductible). Tax rate 30%.

Income statement approach: Year 1 pre tax accounting profit = 10,000 − 7,000 = €3,000. tax expense @30% = €900. Taxable profit = 3,000 + 1,000 (bad debt add back, not yet deductible) = €4,000. current tax expense @30% = €1,200. Deferred tax expense = 900 − 1,200 = −€300 (a deferred tax **asset** increases by 300, since more tax is paid now than the accounting profit implies). Year 2: pre tax accounting profit = 12,000 − 7,500 = €4,500. tax expense @30% = €1,350. Taxable profit = 4,500 − 1,000 (bad debt now deductible) = €3,500. current tax expense @30% = €1,050. Deferred tax expense = 1,350 − 1,050 = +€300 (the deferred tax asset reverses/is used up).

Balance sheet approach cross check: Year 1, accounts receivable GAAP carrying value €8,000 (net of the 1,000 allowance) vs. tax base €9,000 (allowance not recognised for tax) → difference −1,000 → deferred tax asset of 30% × 1,000 = €300, matching the income statement result exactly. Year 2, carrying value €10,000 = tax base €10,000 (both now reflect the write off) → difference zero → deferred tax account back to zero.

Journal entries: Year 1: Dr Tax expense 900. Cr Taxes payable 1,200. Dr DTA (asset) 300. Year 2 (assuming Year 1 taxes payable are settled): Dr Taxes payable 1,200, Cr Cash 1,200

(payment). then Dr Tax expense 1,350, Cr Taxes payable 1,050, Cr DTA 300 (the asset is used up/reversed as the temporary difference closes).

2. Worked Example 2: Accelerated Tax Depreciation Over Six Years

Setup: pre tax accounting profit of €50,000 each year for 5 years, including straight line depreciation of €5,000 (asset costing €25,000, 5 year life, no salvage) and tax exempt municipal bond revenue of €10,000 each year (permanent difference). Tax rate 40%. Tax depreciation follows accelerated rates of 20%, 32%, 19.2%, 11.52%, 11.52%, 5.76% (of the original €25,000 cost) across six years, note tax depreciation runs a year longer than accounting depreciation, at a declining balance style schedule front loaded relative to straight line.

Mechanics, year by year: adjusted accounting profit (after removing the permanent difference) = 50,000 − 10,000 = €40,000 every year (Years 1 to 5). tax expense @40% = €16,000 every year, constant. Tax depreciation in €: Year 1 = 20%×25,000 = 5,000 (matches GAAP exactly, no timing difference yet). Year 2 = 32%×25,000 = 8,000 (exceeds GAAP's 5,000 by 3,000, more deducted for tax now, taxable profit falls, current tax falls, deferred tax liability builds). Year 3 = 19.2%×25,000 = 4,800 (below GAAP's 5,000, timing difference starts reversing). Years 4 and 5 = 11.52%×25,000 = 2,880 each (further below GAAP, reversing further). Year 6 = 5.76%×25,000 = 1,440 (tax depreciation continues into a sixth year, after GAAP depreciation has already fully ended in Year 5, and even generates a small taxable profit adjustment despite no accounting profit that year in the table).

Current tax expense tracks taxable profit each year (Year 1: 16,000. Year 2: 14,800. Year 3: 16,080. Years 4 and 5: 16,848 each. Year 6: an outflow reversal). Deferred tax expense = tax expense (constant 16,000, or 0 in Year 6 since there is no more accounting profit) minus current tax expense, which nets to zero by the end of Year 6 (the deferred tax liability built up in Year 2 is fully unwound by Year 6), confirming the core principle that **temporary differences always fully reverse**, and total tax paid over the asset's life converges to the same amount under both approaches, only the timing differs.

Journal entries pattern (Years 1 to 6): in years where tax depreciation exceeds GAAP depreciation (Year 2), the current tax payable is lower than the accounting tax expense, so the difference is credited to a deferred tax liability (Dr Tax expense 16,000. Cr Taxes payable 14,800. Cr DTA/DTL 1,200). In years where tax depreciation falls below GAAP depreciation (Years 3 to 5), current tax payable exceeds the accounting tax expense, and the previously built liability unwinds, appearing on the debit side of the deferred tax account (Dr Tax expense 16,000. Cr Taxes payable [higher]. Dr DTA/DTL [reversal amount]). By the end of Year 6, the deferred tax account balance returns to zero.

3. Worked Example 3: Balance Sheet Approach, Multiple Items at Once

Bulb Ltd, 30 June 2024: accounts receivable, carrying €150,000 / tax value €175,000. motor vehicles, carrying €165,000 / tax value €125,000. provision for warranty (a liability), carrying €12,000 / tax value €0. deposits received in advance (a liability), carrying €15,000 / tax value €0. development costs (capitalised, an asset), carrying €90,000 / tax value €0 (development costs are tax deductible in the period incurred, i.e. immediately expensed for tax even though capitalised for accounting).

Updated lecture assumption: Use a tax rate of 20% for this example.

Watch out: the extracted slides pose this as a self check exercise ("calculate the temporary differences... justify your classification") without a printed solution. The analysis below applies the balance sheet approach rule from Session 19 directly. verify it against the instructor's own solution.

Applying the balance sheet rule (Session 19, Section 7) to each item:
• Accounts receivable (asset): carrying 150,000 < tax base 175,000 → deductible temporary difference of 25,000 → deferred tax **asset** of 25,000 × 20% = 5,000.
• Motor vehicles (asset): carrying 165,000 > tax base 125,000 (consistent with accounting depreciation at 15% running slower than tax depreciation at 25%) → taxable temporary difference of 40,000 → deferred tax **liability** of 40,000 × 20% = 8,000.
• Provision for warranty (liability): carrying 12,000 > tax base 0 (not deductible until paid) → deductible temporary difference of 12,000 → deferred tax **asset** of 12,000 × 20% = 2,400.
• Deposits received in advance (liability): carrying 15,000 > tax base 0 (already fully taxed when received, so no further tax consequence remains) → deductible temporary difference of 15,000 → deferred tax **asset** of 15,000 × 20% = 3,000.
• Development costs (asset): carrying 90,000 > tax base 0 (already fully deducted for tax when incurred) → taxable temporary difference of 90,000 → deferred tax **liability** of 90,000 × 20% = 18,000.

4. Valuation Allowances

Valuation allowance: a contra account against deferred tax assets. The carrying amount of deferred tax assets is reviewed at the end of each reporting period and reduced if it is no longer probable that sufficient future taxable profit will be available to utilise part or all of the asset.

Must know (valuation allowance mechanics):
• A valuation allowance flows through the income statement as a **loss** when established/increased
• It can be reversed if the profitability outlook improves. a reversal flows through the income statement as a **gain**
• It is **highly discretionary**, based on management's own judgement about future taxable profit, this is exactly the point Section 7 returns to (earnings management potential)

5. How Important Are Deferred Taxes? It Depends on the Country

- The materiality of deferred taxes on a given set of financial statements depends on how independent a country's accounting rules are from its tax rules.
- In the UK, accounting rules operate largely independently of tax rules, producing many temporary differences and a high volume of deferred taxes.
- In Germany and France, accounting rules and tax rules tend to be closely aligned, producing fewer temporary differences, so deferred taxes are comparatively less important there.

6. Why Deferred Taxes Matter for Valuation and Analysis

- Understanding deferred taxes matters for forecasting future cash flows (i.e. for valuation), because it is important to know whether recognised tax assets and liabilities will actually reverse, and therefore have real future cash consequences, or not.
- Financial analysts must examine the components of deferred taxes case by case and judge the likelihood of reversal, considering: future tax rates, tax laws and accounting standards (all of which can change before reversal occurs). the firm's growth rate (a growing firm with rising capital expenditure may perpetually roll forward, rather than reverse, depreciation related deferred tax liabilities). and non recurring items (one off temporary differences behave differently from recurring ones).
- Because valuation allowances involve substantial managerial discretion, they can also be used to smooth reported earnings, a theme the academic research in Section 7 develops further.

7. Academic Research Insights

Must know (four strands of evidence on the book tax gap):
• **Macro predictability:** overall economic activity can predict the book tax gap, and the direction of this relationship depends on whether the economy is expanding or contracting
• **Earnings manipulation / lower earnings quality:** book tax differences are associated with bad prior earnings patterns, financial distress and proximity to bonus thresholds (Mills and Newberry 2001). firms with higher deferred tax expense are more likely to avoid earnings declines and losses (Phillips et al. 2003). larger book tax differences are associated with less persistent earnings, and investors treat large positive book tax differences as a "red flag," reducing their expectation of future earnings persistence (Hanlon 2005)
• **Value relevance:** the ratio of taxable income to book income helps predict subsequent five year earnings growth (Lev and Nissim 2004)
• **Book tax conformity reduces informativeness:** Hanlon et al. (2008) studied firms forced to switch from the cash method to the accrual method for tax purposes (which narrows the book tax gap) and found this increase in book tax conformity **reduced** the informativeness of reported accounting earnings, i.e. a wider book tax gap is not purely a red flag, it can also reflect informative, independent accounting judgement

- The session closes with a real world illustration referenced from the Financial Times (June 2018): "Berkshire Hathaway's $62bn Secret: Interest Free Government Loan?", a pointed way of framing deferred tax liabilities as, in substance, an interest free loan from the government for as long as the underlying temporary differences keep rolling forward rather than reversing

(consistent with Section 6's growth rate point).

8. Common Mistakes and Exam Traps

- Assuming a deferred tax liability that keeps growing (e.g. from ongoing capex outpacing tax depreciation reversal) will eventually cause a cash outflow soon, for a growing firm, such liabilities can persist and even grow indefinitely without reversing, which is exactly the "interest free loan" framing.
- Forgetting that a valuation allowance affects only **deferred tax assets**, not liabilities, it exists because a DTA's benefit depends on having enough future taxable profit to use it against. there is no equivalent judgement call for a DTL, which always represents a real future obligation.
- Treating a large book tax gap as automatically a sign of manipulation, the academic evidence is mixed: it is associated with lower earnings quality in some studies (Hanlon 2005, Mills and Newberry 2001), but conversely, forcing the gap to narrow (higher book tax conformity) has been shown to reduce earnings informativeness (Hanlon et al. 2008).
- Missing that accelerated tax depreciation schedules (like the 20/32/19.2/11.52/11.52/5.76% pattern in Example 2) mean the tax life can differ in **length**, not just in early/late weighting, from the accounting useful life, the tax depreciation ran for six years here against a five year accounting life.
- Confusing which side of the balance sheet approach table each item in Example 3 falls on, always determine asset vs. liability status first, since the direction rule (Session 19, Section 7) is mirror imaged between the two.

9. Must Know Absolutely

Must know:
• Multi year deferred tax always fully reverses by the end of the underlying temporary difference's life, total tax paid converges regardless of timing
• Journal pattern: Dr Tax expense (accounting profit based) / Cr Taxes payable (taxable profit based) / plug the difference to DTA (debit) or DTL (credit)
• Valuation allowance: contra account reducing a DTA when future taxable profit is no longer probable. flows through P&L (loss to establish, gain to reverse). highly judgement based
• Deferred taxes matter more where accounting and tax rules diverge (e.g. UK) than where they align (e.g. Germany, France)
• For valuation: distinguish DT balances likely to reverse (real future cash effect) from those likely to keep rolling forward (growing firms, capex driven DTLs)
• Book tax gap evidence is two sided: associated with lower earnings quality/manipulation risk in some studies, but forcing conformity (narrowing the gap) has also been shown to reduce earnings informativeness

Watch Out

Watch out: Example 3 (Bulb Ltd) has no printed solution in the extracted slides, Section 3's analysis is a self constructed application of the Session 19 balance sheet rule, not a verified answer key. confirm it against the instructor's solution.